

Report
Of
Generative AI Applications in Financial Data Analysis and
Development of an AI-Powered Financial Chatbot

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School of Computer Science and Engineering
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In partial fulfilment of the requirement of degree of
B.Tech CSE



By
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CANDIDATE'S DECLARATION

I, **ROHAN SINGH**, do hereby declare that the internship / job simulation report titled "***Generative AI Applications in Financial Data Analysis and Development of an AI-Powered Financial Chatbot***" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted, and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to IILM University and the School of Computer Science and Engineering for providing me with the guidance and academic framework necessary to undertake and document this Generative AI (GenAI) Job Simulation as part of my internship requirement.

I am thankful to my faculty guide, [Faculty Name], and the faculty members of the department for their valuable guidance, constructive feedback, and continuous encouragement throughout the preparation of this report.

I would also like to acknowledge Forage for providing the platform through which this industry-oriented virtual job simulation, developed in association with BCG X, was made available. This simulation gave me practical, self-paced exposure to real-world tasks in data extraction, financial data analysis, and the development of an AI-powered financial chatbot, and helped me understand how Generative AI concepts are applied in a professional, business-oriented context.

Finally, I extend my heartfelt thanks to my family, friends, and peers for their constant support, encouragement, and motivation throughout this endeavour.

Signature: _____

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INTERNSHIP / JOB SIMULATION COMPLETION CERTIFICATE

The certificate presented below was issued by Forage on completion of the BCG X GenAI Job Simulation, confirming the completion of practical tasks in data extraction and initial analysis, and in developing an AI-powered financial chatbot. This document is presented as evidence of the industry-oriented virtual experience undertaken and reported in this document.

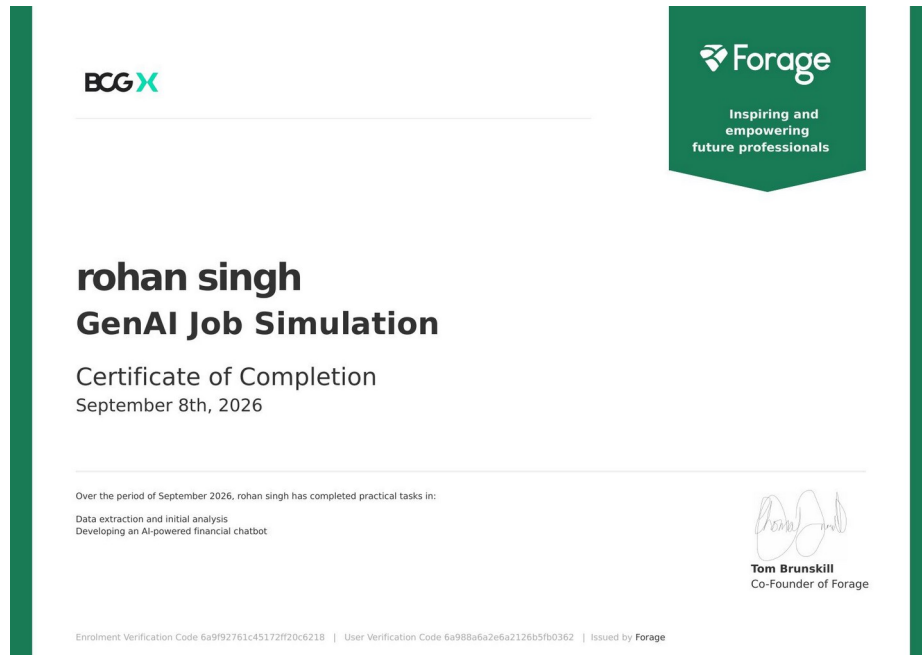


Figure: Certificate of Completion – GenAI Job Simulation (Forage / BCG X)

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CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

Generative Artificial Intelligence (GenAI) refers to a class of artificial intelligence systems capable of producing new content — such as text, summaries, explanations, and conversational responses — by learning patterns from large volumes of existing data. Unlike traditional rule-based software, GenAI systems, including large language models, can interpret natural language queries and generate contextually relevant responses, which makes them particularly useful in domains that require interaction between humans and complex data.

In recent years, GenAI has become increasingly relevant to modern software engineering and business operations. Organizations are exploring how such systems can automate repetitive analytical tasks, support decision-making, and make specialized information more accessible to non-technical users. The finance sector, in particular, generates large volumes of structured and semi-structured data — such as financial statements, filings, and reports — that require extraction, cleaning, and analysis before they can be used effectively. GenAI offers the potential to streamline this process and present the resulting insights through natural, conversational interfaces.

Within this broader context, this report is based on a Generative AI Job Simulation completed on the Forage platform, developed in association with BCG X. As confirmed by the Certificate of Completion (September 8th, 2026), the simulation involved practical tasks in two key areas: (i) data extraction and initial analysis, and (ii) developing an AI-powered financial chatbot — a realistic workflow beginning with raw financial information and ending with a conversational tool that communicates insights derived from it. Raw financial data is rarely usable in its original form; it must be identified, extracted, cleaned, and organized before meaningful analysis, after which the chatbot serves as an accessible interface for querying that information in natural language. This job simulation gave practical, self-paced exposure to these concepts through guided tasks and example solutions, and this report documents that experience in the university's prescribed format.

4.2 Organization Profile

This section describes the platforms and organizations associated with the job simulation on which this report is based, strictly to the extent supported by the certificate and by publicly available information.

Forage is an online platform that provides free, self-paced "virtual job simulations" in partnership with a range of companies. These simulations are designed to give students and early-career professionals practical, industry-style exposure to the kind of tasks performed within a given company or role, without requiring a formal application process or an employment relationship. Participants typically work through a series of guided tasks, supported by pre-recorded instructional videos and example/reference solutions, and receive a Certificate of Completion upon finishing the program.

BCG X is described publicly as the technology build and design unit of Boston Consulting Group (BCG), combining consulting expertise with capabilities in data science, software engineering, and artificial intelligence to help organizations address complex, technology-oriented business challenges. The GenAI Job Simulation referenced in this report was developed in association with BCG X and made available through the Forage platform, framing the simulation's tasks around a realistic, BCG X–style business scenario involving financial data and Generative AI.

It is important to state clearly that this job simulation does not constitute an employment or internship relationship between the participant, Forage, or BCG X. It is a virtual, industry-oriented practical exercise intended for learning and skill development. Accordingly, this report describes the experience using terms such as "job simulation," "virtual experience," and "industry-oriented practical exercise," rather than characterising it as a traditional workplace internship.

Such simulations are relevant to professional skill development because they expose participants to realistic problem framings, industry terminology, and example-based evaluation of their work, which supports the development of applied technical and analytical skills that complement classroom learning.

4.3 Problem Statement

Financial information — whether in the form of statements, filings, or reports — is often extensive, technical, and difficult for non-specialist users to interpret quickly. Extracting the relevant figures from such sources, organizing them into a usable structure, and performing an initial analysis to identify meaningful patterns is a necessary but time-consuming step before that information can support any decision-making. Furthermore, once financial information has been analyzed, it is not always presented in a way that is easily accessible to end users, who may not have the technical background to interpret raw data or complex reports directly.

The problem addressed by this job simulation, therefore, is two-fold: first, how to extract useful information from financial data sources and convert it into a form suitable for analysis; and second, how to make the resulting financial information accessible to users through an intuitive, conversational interface powered by Generative AI, so that users can interact with and query financial information more naturally than through manual review of raw data.

Consistent with the scope of the certificate, this report does not reference any specific proprietary dataset, company-specific financial figures, or client business requirements beyond what is generally associated with this category of exercise, since such details are not confirmed by the source certificate.

Framed this way, the problem is representative of a broader class of challenges faced across the finance and consulting industry, where the volume and technical density of financial data routinely outpaces the ability of non-specialist stakeholders to interpret it without assistance, making this an appropriate and realistic problem for a GenAI-focused job simulation to address.

4.4 Project Objectives

The objectives of this job simulation were framed around the two confirmed task areas — data extraction/initial analysis and the development of an AI-powered financial chatbot — and were intended to build both technical familiarity and broader professional skills. The key objectives are summarized below:

- To understand the practical applications of Generative AI in a business and financial context.
- To gain hands-on exposure to the process of data extraction and initial analysis of financial information.
- To understand how raw financial data can be processed and organized into meaningful insights.
- To understand, at a conceptual level, the design of an AI-powered financial chatbot capable of responding to financial queries.
- To improve analytical thinking and problem-solving skills through a structured, industry-style exercise.
- To understand how Generative AI can improve accessibility to financial information for non-specialist users.
- To gain exposure to industry-oriented GenAI workflows as practised in a simulated business consulting environment.

Collectively, these objectives were designed to move the participant from a conceptual understanding of Generative AI toward a practical appreciation of how such technology can be applied to a real, business-relevant problem within the finance domain.

4.5 Scope of the Project

The scope of this project, as reflected in the GenAI Job Simulation, covers the following areas:

- Extraction of relevant information from financial data sources.
- Initial-level analysis of the extracted financial data to identify patterns or notable figures.
- Conceptual and practical exposure to AI-assisted processing of financial information.
- Design-level understanding of conversational interaction with financial information through a chatbot interface.
- Discussion of potential real-world applications of AI-powered financial chatbots.
- Consideration of future possibilities for extending such tools with more advanced Generative AI capabilities.

This project is subject to the following limitations, which are stated explicitly to keep the report's claims proportional to the evidence available:

- This work was completed as a virtual job simulation and not as a production-grade or commercially deployed system; no claim of real-world deployment is made.
- The financial chatbot discussed in this report is a learning exercise and must not be treated as a substitute for professional financial advice.
- The reliability and usefulness of any AI-powered financial chatbot depends heavily on the quality of the underlying data and the reliability of the underlying model, both of which have inherent limitations.

4.6 Technologies and Tools Used

The Certificate of Completion issued for this GenAI Job Simulation confirms the two task areas completed — data extraction and initial analysis, and development of an AI-powered financial chatbot — but does not itself specify the exact programming languages, APIs, frameworks, databases, language models, or cloud platforms used. To avoid overstating what can be verified from the certificate, this section describes the broad categories of technology relevant to the simulation rather than asserting specific tools as confirmed personal usage.

Category	Relevance to the Job Simulation
Generative AI	Used conceptually to interpret user queries and generate natural-language responses about financial information.
Data Extraction	Used to identify and pull relevant figures/fields from financial data sources for further processing.
Data Analysis	Used to perform initial, exploratory-level analysis of the extracted financial data.
Conversational AI	Used to structure the chatbot's interaction model so it can respond to financial queries.
Financial Information Processing	Used to organize processed financial data into a form suitable for chatbot-based retrieval.

Table 4.1: Categories of Technologies and Tools Involved in the GenAI Job Simulation

For context only, publicly available descriptions of similar BCG X GenAI Job Simulations on Forage indicate that participants commonly use general-purpose data analysis tools and simple, rule-based conversational logic to build a prototype chatbot. This is background context from public program descriptions, not a confirmed record of the tools used in this instance, since the certificate itself does not list them.

4.7 System Architecture

Since the certificate does not provide an explicit system architecture diagram, the following is presented as a conceptual architecture, illustrating a logical flow consistent with the two task areas described on the certificate: data extraction/initial analysis, and the AI-powered financial chatbot.

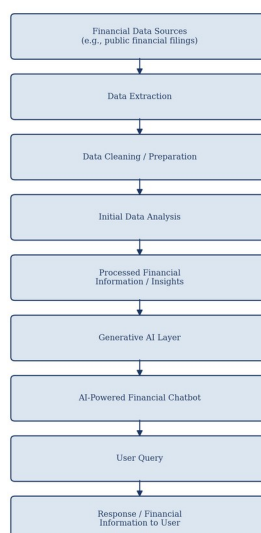


Figure 4.1: Conceptual Architecture of the AI-Powered Financial Chatbot

Component explanation:

- Financial Data Sources — publicly available or provided financial information such as statements or filings.
- Data Extraction — identifying and pulling relevant fields/figures from the sources.
- Data Cleaning / Preparation — organizing and correcting the extracted data.
- Initial Data Analysis — exploratory analysis to identify patterns or notable figures.
- Processed Financial Information — the structured output ready for the chatbot layer.
- Generative AI Layer — interprets natural-language queries and generates responses.
- AI-Powered Financial Chatbot — the user-facing conversational application.
- User Query and Response — the interaction cycle between user and chatbot.

4.8 Methodology

The methodology followed during the GenAI Job Simulation can be summarized as a sequence of logical phases, consistent with the two task areas confirmed by the certificate:

Phase	Description
1	Understanding the business/financial problem and the objectives of the simulation.
2	Extracting relevant financial data from the given source(s).
3	Performing initial data analysis on the extracted financial data.
4	Identifying useful financial information and patterns from the analysis.
5	Designing the intended interaction flow for the financial chatbot.
6	Applying Generative AI concepts to enable natural-language interaction with the processed financial information.
7	Reviewing the chatbot's responses for clarity and relevance to the query asked.
8	Documenting the process, outcomes, and learnings in this report.

Table 4.2: Methodology Adopted During the GenAI Job Simulation

As no specific quantitative evaluation metrics are documented on the certificate, this report does not claim any particular accuracy figures, performance benchmarks, or evaluation scores for the chatbot; the review in Phase 7 refers to a general, qualitative check of response clarity and relevance, consistent with the exploratory and educational nature of the simulation.

4.9 Expected Outcomes

The outcomes of this job simulation can be separated into learning outcomes and technical/project outcomes, as follows:

Learning Outcomes:

- Better conceptual understanding of financial data and the steps required to make it usable.
- Practical exposure to the process of data extraction from real or realistic data sources.
- Understanding of how initial-level data analysis is carried out and why it matters.
- Improved understanding of Generative AI and its applications beyond generic text generation.
- Familiarity with the principles of conversational AI and chatbot design.
- Improved analytical thinking and structured problem-solving ability.
- Industry-oriented practical learning through a realistic, guided business scenario.

Technical / Project Outcomes:

- Conceptual familiarity with a pipeline connecting financial data extraction, analysis, and chatbot-based delivery of insights.
- Better understanding of how AI-powered financial assistants can be structured at a design level.
- Awareness of the practical considerations and limitations involved in building AI tools for financial information delivery.

4.10 Certificates of Completion and Other Communication Proof

The Certificate of Completion for the GenAI Job Simulation, issued by Forage and confirming completion of tasks in data extraction and initial analysis, and in developing an AI-powered financial chatbot, has been included earlier in this report (see the Internship / Job Simulation Completion Certificate section on page iii).

The certificate records the participant's name (Rohan Singh), the program name (GenAI Job Simulation), the date of completion (September 8th, 2026), and is signed by Tom Brunskill, Co-Founder of Forage. It also carries a unique Enrolment Verification Code and User Verification Code, which allow the authenticity of the certificate to be independently verified through Forage's platform.

No communication mail proof (such as onboarding emails or task-submission acknowledgements) was provided in the available source documents. Accordingly, in line with the requirement to avoid fabricating evidence, no such correspondence has been reproduced or referenced in this report.

Together, the certificate and its verification codes serve as the primary documentary evidence for the completion of this job simulation and the activities described throughout this chapter.

CHAPTER 5

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